

Correlation Matrix Computation

1 Introduction

This document illustrates how to compute the correlation matrix for given segments of data and how to average these matrices over multiple segments. Each segment consists of 32 elements, alternating between two series A and B .

2 Segment Definition

Each segment is defined as:

$$\text{Segment} = [A_1, B_1, A_2, B_2, \dots, A_{16}, B_{16}]$$

3 Correlation Matrix Computation

The correlation matrix C is an 8x8 matrix where each element C_{ij} is computed using the elements from the segment. Specifically, the element C_{ij} is given by:

$$C_{ij} = (A_{i+1} - A_{i+9})(B_{j+1} - B_{j+9})$$

for $i, j \in \{0, 1, \dots, 7\}$.

3.1 Matrix Elements

For example:

$$\begin{aligned} C_{00} &= (A_1 - A_9)(B_1 - B_9) \\ C_{01} &= (A_1 - A_9)(B_2 - B_{10}) \\ C_{10} &= (A_2 - A_{10})(B_1 - B_9) \\ &\vdots \\ C_{77} &= (A_8 - A_{16})(B_8 - B_{16}) \end{aligned}$$

4 Averaging the Correlation Matrices

Given N segments, each producing a correlation matrix C_k , the final average correlation matrix $\bar{C}$ is calculated by averaging each corresponding element across all segment matrices:

$$\bar{C}_{ij} = \frac{1}{N} \sum_{k=1}^N C_{ij}^{(k)}$$

where $C_{ij}^{(k)}$ is the (i, j) -th element of the k -th segment's correlation matrix and N is the total number of segments.

4.1 Example

For $N = 3$ segments, the average matrix $\bar{C}$ would be:

$$\bar{C}_{ij} = \frac{1}{3} \left(C_{ij}^{(1)} + C_{ij}^{(2)} + C_{ij}^{(3)} \right)$$

5 Summary

The formula for each element of the correlation matrix is:

$$C_{ij} = (A_{i+1} - A_{i+9})(B_{j+1} - B_{j+9})$$

The final average correlation matrix is:

$$\bar{C}_{ij} = \frac{1}{N} \sum_{k=1}^N C_{ij}^{(k)}$$

6 Aggregated Correlation Matrix

The correlation matrices for all segments are arranged into an aggregated correlation matrix M , where each column represents a correlation matrix of one segment:

$$M = \begin{bmatrix} C_{00}^{(1)} & C_{00}^{(2)} & \cdots & C_{00}^{(N)} \\ C_{01}^{(1)} & C_{01}^{(2)} & \cdots & C_{01}^{(N)} \\ \vdots & \vdots & \ddots & \vdots \\ C_{77}^{(1)} & C_{77}^{(2)} & \cdots & C_{77}^{(N)} \end{bmatrix}$$

7 Vector for Averaging

The vector $\mathbf{v}$ used for averaging is:

$$\mathbf{v} = \begin{bmatrix} \frac{1}{N} \\ \frac{1}{N} \\ \vdots \\ \frac{1}{N} \end{bmatrix}$$

8 Matrix-Vector Multiplication

The resulting averaged correlation matrix $\bar{C}$ is obtained by multiplying the aggregated correlation matrix M by the vector $\mathbf{v}$:

$$\bar{C} = M\mathbf{v}$$

This can be illustrated as:

$$\bar{C} = \begin{bmatrix} C_{00}^{(1)} & C_{00}^{(2)} & \cdots & C_{00}^{(N)} \\ C_{01}^{(1)} & C_{01}^{(2)} & \cdots & C_{01}^{(N)} \\ \vdots & \vdots & \ddots & \vdots \\ C_{77}^{(1)} & C_{77}^{(2)} & \cdots & C_{77}^{(N)} \end{bmatrix} \begin{bmatrix} \frac{1}{N} \\ \frac{1}{N} \\ \vdots \\ \frac{1}{N} \end{bmatrix}$$

9 Resulting Averaged Correlation Matrix

Each element of the final averaged correlation matrix $\bar{C}$ is:

$$\bar{C}_{ij} = \sum_{k=1}^N \frac{1}{N} C_{ij}^{(k)}$$